X3 / MAINNET-READINESS AUDIT ARTIFACT

X3 ATOMIC STAR

THE ROAD TO MAINNET

Current-state refresh at `master`

Commit:	`d082f18de`
Refresh date:	2026-09-08
Predecessor:	2026-09-05 field manual frozen at `6a24d8cf`
Generator:	Typst `X3-CURRENT-STATE-2026-09-08.typ`

PUBLIC TESTNET: NO-GO
MAINNET: NO-GO

This document supersedes the Sep 5 cover claim for current-master evidence only. The historical 127-page field manual remains a frozen artifact for commit `6a24d8cf`. This refresh is not an independent security certification.

What changed since the Sep 5 field manual

The Sep 5 manual was correct for its audited source state but is no longer current. Since then the workspace moved from blocked builds and unverified atomic paths to passing local gates plus an implemented and tested atomic lifecycle. The most important current facts:

- Workspace Rust check, full tests, and clippy pass with the repository's dev-node evidence path.
- The JavaScript test and production-build suites pass; the Python root test suite passes (181 tests).
- `cargo audit` reports 0 blocking findings.
- GitHub Dependabot critical alerts are 0 after PR 124 remaining alerts are 41 high / 85 medium / 27 low.
- The atomic-kernel overlay diff/reverter, `x3leg:` receipt path, `x3fin:` finalization, and forced rollback control are present and covered by workspace tests.
- The clean embedded runtime build now works from an empty nested target without `SKIP_WASM_BUILD`: the pinned toolchain adds `wasm32v1-none`, the vendored `sp-wasm-interface` no longer leaks `parity-scale-codec/std`, and `x3-cross-vm-bridge` std-only crypto (`blst`, `ed25519-dalek`) is optional and std-gated. `scripts/ci/check_runtime_wasm_non_stub.sh` passes on the release artifact.
- Three of the four Critical findings in the 2026-09-06 independent audit are closed in current source: fabricated wallet RPC is dev-only, `report_misbehavior`/`slash_bond` require root, and the readiness registry now has a real-test consistency check.
- HIGH-API-1 is closed in source: the gateway `/health` and `/readyz` endpoints run a real DB round-trip and return 503 when it fails, `/livez` preserves process liveness, and the registry no longer cites fabricated `/health/<feature>` paths.
- HIGH-TX-2 is closed in source: `x3_submitCrossVmTransaction` is registered only on Development/Local chain specs, no longer auto-submits wrapped register/mint under a threshold-1 council call, and every Live chain spec requires at least two council members.
- HIGH-CK-2 is closed in source: the simulated primitives are renamed `kyber_research` and `dilithium_research`, and crate/package docs explicitly forbid real-security use until audited PQC bindings are integrated.
- HIGH-VM-1 is closed in source: `x3_htlc` is added to the SVM workspace with six HTLC-specific unit tests (plus Anchor's `test_id`) covering timelock bounds, preimage hashlock checks, expiry, and PDA derivation.
- A fresh four-validator reserved full-mesh drill converged to one finalized chain on cold start and survived loss of each individual validator with the remaining three finalizing one chain.
- The fourth Critical, recoverable secrets in git history, remains open.

Fresh evidence ledger

Command / check	Exit	Observed result
`git status` / `git log -1`	0	`master` clean at `d082f18de`
`cargo check --workspace --locked` with `SKIP_WASM_BUILD=1`	0	Workspace compiles; cached runtime WASM embedded
`cargo test --workspace` with `SKIP_WASM_BUILD=1`	0	All active suites pass; 3 known manual node tests ignored
`cargo clippy --workspace --all-targets -- -D warnings` with `SKIP_WASM_BUILD=1`	0	No workspace warnings as errors
`cargo build --release -p x3-chain-runtime` with `SKIP_WASM_BUILD` unset and empty nested target	0	WASM rebuilt from source with `wasm32v1-none`; real compact/compressed blob emitted
`bash scripts/ci/check_runtime_wasm_non_stub.sh`	0	No stubbed `wasm_binary.rs`; non-trivial WASM artifact present
`cargo test -p x3-gateway router_serves_real_http_over_ephemeral_port_without_db`	0	`/livez` 200; `/health` and `/readyz` 503 with `db:false` against a dead pool
`cargo test --manifest-path X3-contracts/svm/Cargo.toml -p x3_htlc`	0	7 HTLC tests pass (timelock, hashlock, expiry, PDAs)
`cargo test -p quantum-crypto`	0	22 tests pass with renamed research/simulated modules
`cargo test -p x3-chain-node --lib`	0	48 pass; 3 ignored; live-node RPC integration included
`TESTNET_BASE=/tmp/x3-mesh-4 python3 scripts/testnet/run-mesh.py cycles --count 4 --cycles 1`	0	Fresh four-validator full mesh converged in 14s with one finalized head
`TESTNET_BASE=/tmp/x3-mesh-4 python3 scripts/testnet/run-mesh.py kills --count 4`	0	4/4 single-validator losses left 3/3 survivors finalizing one common chain
`cargo audit`	0	0 blocking; 27 allow-listed warnings
`npm test`	0	All configured JS test packages pass
`npm run build`	0	All configured JS builds pass
`pnpm audit` in `apps/x3-studio`	0	0 vulnerabilities
`.venv/bin/python -m pytest tests -q`	0	181 passed
GitHub open critical Dependabot alerts	0	Verified after PR 124 merge

Relevant passing Rust suites observed in the workspace test run include:

- `cross_vm_real_chain_test.rs`: node binary, RPC methods, WebSocket connection, signed extrinsic submission, block production, and finality.
- `live_internal_mainnet_e2e.rs`: timeout expiry, duplicate/reordered delivery, bridge-proof crypto, and full accounting paths.
- `atomic_swap_orchestrator`: receipt-root commitment, deterministic bundle IDs, and finalization request construction.
- `x3-bridge-adapters`: overlay diff round-trips with the atomic-kernel format and runtime-dispatcher execution through the client-backed escrow config.
- `pallet-x3-cross-vm-router` and supply-ledger invariant tests.

Current finding status

The table below carries forward the 2026-09-06 independent audit's register (`fbd4613b`) and updates it only where current source and this session's checks support a status change. Unchanged rows remain open unless a new commit explicitly closes them; none are silently deleted.

Finding	Status in refresh	Basis
CRITICAL-TX-1	Closed in source	`wallet_*` RPC registration is gated by `ChainType::Development` in `node/src/rpc.rs:890` and `node/src/service.rs:1054`
CRITICAL-CN-1	Closed in source	`report_misbehavior` and `slash_bond` enforce `ensure_root`; tests cover root-only calls
CRITICAL-TOK-1	Closed in source	`scripts/check-readiness-consistency.sh` now verifies `required_tests` against real functions; registry fiction was purged
CRITICAL-CK-1	Open	`git log --all` still returns commits for `sepolia-deployer-wallet.txt` and validator seed summaries; history rewrite + rotation remains required
HIGH-TX-2	Closed in source	`x3_submitCrossVmTransaction` is Development/Local-only and no longer auto-mints wrapped assets; Live chain specs require at least two council members
HIGH-CN-2	Closed in source	`LAUNCH_SCOPE.md` now explicitly lists permissionless validator staking/bonding/nomination as NOT IMPLEMENTED and deferred to M3
HIGH-CK-2	Closed in source	Simulated modules renamed `kyber_research`/`dilithium_research`; docs and Cargo description forbid production use
HIGH-VM-1	Closed in source	Six HTLC unit tests plus Anchor `test_id` pass in the SVM workspace
HIGH-API-1	Closed in source	`/health` and `/readyz` run a real DB round-trip and return 503 when down; `/livez` is liveness; registry `health_endpoint` fiction removed
MEDIUM / LOW / INFO rows	Open unless code inspection confirms closure	Carried from the 2026-09-06 register

Open Critical:

The only remaining Critical carried into this refresh is CRITICAL-CK-1. Because genuine validator and deployer secrets are recoverable from git history, this alone keeps public testnet and mainnet at NO-GO until history is rewritten and all affected keys are rotated offline.

Current scorecard

This refresh replaces the old single-number readiness display with a small evidence scorecard. It does not invent a new aggregate percentage because no complete 64-feature independent re-audit was performed in this session. Each row is a concrete, current claim.

Criterion	Status	Evidence / limit
Workspace Rust build	PASS	`cargo check --workspace --locked` with `SKIP_WASM_BUILD=1`
Rust tests	PASS	`cargo test --workspace`; active suites pass, manual node tests ignored
Rust lint	PASS	`cargo clippy --workspace --all-targets -- -D warnings`
Rust dependency advisory	PASS	`cargo audit` exit 0; 0 blocking
JS tests / builds	PASS	`npm test`, `npm run build`, pnpm studio audit
Python tests	PASS	181 passed
GitHub critical advisories	PASS	0 critical after PR 124
Atomic lifecycle integration	PARTIAL PASS	Overlay, receipt, finalization, rollback, and live node tests pass; no independent multi-validator review
Clean from-scratch WASM compile	BLOCKED	Nested duplicate-core failure with `SKIP_WASM_BUILD` unset; cached-WASM workaround used
Git history secret purge	FAIL	CRITICAL-CK-1 remains open
Multi-node finality / recovery drill	NOT VERIFIED	No fresh four-validator independent run in this refresh
Independent security audit	NOT VERIFIED	Not commissioned
External bridge enabled	NOT APPLICABLE / DISABLED	Genesis governance flag keeps external routes disabled

Decisions and next steps

Public testnet: NO-GO. Mainnet: NO-GO.

The NO-GO is not because the workspace fails to build; the local gates are now green. It is because a current Critical finding (recoverable secrets in git history) remains open and because public operation still needs independent security review and multi-validator operational proof.

Next steps in priority order:

1. Rewrite git history and rotate every affected key offline; verify `git log --all` no longer returns the secret paths.
2. Engage an independent professional security audit of the runtime and contract stacks.
3. Close the remaining High findings, starting with unilateral wrapped-asset authority and the fake post-quantum placeholder naming.
4. Run a fresh four-validator finality/partition/recovery drill with retained raw evidence.
5. Repair the clean embedded WASM build without the cached-artifact workaround and promote it to a gate.
6. Refresh the full 64-feature scorecard only after the independent audit and multi-node drill provide current evidence for each row.

Regeneration

Compile this PDF from the artifact directory with:

```
typst compile --root . X3-CURRENT-STATE-2026-09-08.typ X3-ROAD-TO-MAINNET.pdf
```

The historical audit files remain beside this source. Checksums and manifest entries should be regenerated after this PDF is published.